HOANG CAO PHONG

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EDUCATION

Vietnam National University, Ho Chi Minh City – University of Science **Sep 2024 – Jun 2028**

- B.Sc. in IT| **Specialization:** Computer Science – Artificial Intelligence Track *(expected)*
- **Selected Coursework:** AI Fundamentals, Machine Learning, NLP, Computer Vision, Data Mining, Smart Data Analytics, Scientific Method, Linear Algebra, Calculus, Big Data, Deep Learning, Speech and Video Processing.
- ***Honor:*** *Selected for University ICPC Team – University of Science, VNU-HCM (2024)*

Tran Hung Dao High School for the Gifted, Phan Thiet City, Binh Thuan, Vietnam **Sep 2021 – Jun 2024**

- Chemistry Gifted Class
- ***Honor:*** Selected as 1 of 6 outstanding 8th-grade students from Binh Thuan Province chosen for Chemistry Team in **Talent Class for 9th Graders** at *Tran Hung Dao High school for the Gifted.*

AWARDS & CERTIFICATIONS

- Outstanding Trainer Award - Gemini Academy for Student **Jun 2026**
- Participant - The 3rd Youth Science Conference 2026 - University of Science, VNU-HCM **Mar 2026**
- Applying AI in Learning and Research - Issued by Student Culture House of Ho Chi Minh City **Apr 2025**
- Medical Imaging workshop - Certified by Kyoto Institute of Technology, Japan **Feb 2025**
- Robot Programming - Certified by United States of America Consulate General – Ho Chi Minh City **Oct 2024**
- Third Prize - Provincial Informatics Olympiad **Oct 2023**

RESEARCH EXPERIENCE

UIT Information Security Lab (UIT - InSecLab) | AI x Security Research Intern **Mar 2026 – Ongoing**

- Led a small research group on AI-assisted cybersecurity, focusing on security analysis and multimodal AI.
- Supported research workflows, including literature review, data preprocessing, model evaluation, and result analysis.
- Assisted in developing and maintaining web-based resources for the lab and CNSC Information Security Center.

Computational Linguistics Center – HCMUS | Research Assistant Intern **Jan 2026 – Ongoing**

- Selected as a Research Assistant Intern at an interdisciplinary research center specializing in Linguistics, Computer Science, and Natural Language Processing (NLP), with a focus on Vietnamese-related resources.
- Supported Vietnamese language resource development, including data preprocessing and linguistic annotation.
- Collaborated in academic research workflows, gaining exposure to applied AI and NLP research practices.

TECHNICAL EXPERIENCE & ACTIVITIES

Google Student Ambassador Program | Google Student Ambassador 2026 **Jan 2026 – Ongoing**

- Spearheaded campus initiatives to advocate for Google's AI and developer ecosystem.
- Orchestrated technical workshops equipping students with practical insights into AI trends.
- Cultivated a dynamic tech community to accelerate cutting-edge technology adoption.

Google Developer Group on Campus – HCMUS | MC & Graphic Designer **Oct 2024 – Ongoing**

- Served as MC for technical workshops and community events, introducing sessions and coordinating program flow.
- Designed visuals for events, workshops, and social media using Illustrator, Photoshop, GIMP, and Premiere.
- Collaborated with developers and organizers to align technical presentations with branding and audience engagement

Gemini Academy for Student | Trainer **Sep 2025 – Ongoing**

- Delivered student-focused training sessions on Google Gemini and AI learning applications.
- Supported participants in understanding practical use cases of generative AI for study, productivity, and research.
- Recognized for excellent completion of the Trainer role in the Gemini Academy for Student program..

Library of University of Science – VNUHCM | Member of Technical Department **Oct 2024 – Ongoing**

- Set up audiovisual systems and online streaming for library events.
- Produced videos, newsletters, and social media updates to boost student engagement.
- Conducted data analysis and research to improve library services.

Strong PC – Student Laptop Service & Optimization | Co-founder, Management Board **Sep 2025 – Ongoing**

- Co-founded and led a student-run laptop support project providing maintenance and optimization services.
- Managed operations, task assignments, and financial planning across technical and communication teams.
- Directed internal documentation, workflow development, and collaboration planning with the University Library

TECHNICAL PROJECTS

WindyAI – Smart Travel Planner | Project Manager & AI Engineer **Sep 2025 - Dec 2025**

- Formulated **travel itinerary planning** as a **constrained optimization problem** *(time, budget, spatial, preference constraints)*, and designed a **Greedy + Lookahead heuristic** with a **multi-factor POI scoring function** to generate feasible, near-optimal travel sequences under real-world constraints.
- Developed an **image-based landmark recognition module** using a **ResNet-based CNN (PyTorch)** on user-uploaded photos, and integrated visual predictions into the itinerary planning pipeline to reduce ambiguity in POI selection.
- Built an **end-to-end computer vision inference pipeline** *(preprocessing, model inference, top-K prediction with confidence scores)* and tightly coupled recognition outputs with downstream planning decisions.
- **Led a team of five** in designing an **end-to-end AI decision support system**, integrating **auxiliary AI modules** *(weather-aware adjustment, LLM-based assistant)*, and exploring reinforcement learning–based planning.

***Link Project*:** https://github.com/HoangCaoPhong/WindyAI-Smart_Travel_Planner

Vietnamese Facial Emotion Recognition | Computer Vision, Deep Learning **Dec 2025 - Ongoing**

- Developed a **real-time facial emotion recognition system** using **CNNs and OpenCV**, and formulated the task as an evaluation of cross-cultural generalization in facial expression recognition.
- Analyzed **dataset bias** of **FER2013 (Western-centric)** on Vietnamese facial data, identifying performance degradation and misclassification patterns across emotion categories.
- Designed and implemented a **transfer learning and fine-tuning pipeline** with Vietnamese facial images, applying **layer freezing** and **low learning-rate optimization** to adapt pre-trained models under limited data conditions.
- Evaluated model performance using **accuracy and confusion matrices**, and analyzed **limitations related to dataset size and cultural expression variance**, highlighting challenges in emotion recognition across cultural contexts.

***Link Project*:** *https://github.com/HoangCaoPhong/realtime-facial-emotion-recognition-vietnamese-faces*

MindMeetingAI – Intelligent Meeting Assistant | AI Engineer, NLP & LLM Systems **Sep 2025 – Dec 2025**

- Designed an **end-to-end NLP system** for meeting speech transcription, structured summarization, and semantic question answering, framing the task as an applied study of **information retrieval and generation over speech data**.
- Implemented **Vietnamese ASR** using **Whisper**, and built a **transcript normalization pipeline** to improve downstream retrieval and summarization quality.
- Designed a **Retrieval-Augmented Generation (RAG) pipeline** using **sentence embeddings** and a **vector database**, enabling context-aware semantic search over long meeting transcripts.
- Integrated **local LLM inference** for privacy-preserving analysis, and analyzed **RAG effectiveness and failure cases** in summarization and semantic Q&A over noisy ASR outputs.

Keywords: *Speech-to-Text · RAG · LLM Systems · Semantic Search · Sentence Transformers · Vector Databases*

***Link Project*:** *https://github.com/HoangCaoPhong/MindMeetingAI*

MAYBAO, E-commerce clothing website | Team Lead , Web Developer **Sep 2024 – Dec 2024**

- Led a team of four, managed tasks using Scrum methodology & Notion, earning a perfect score for all members.
- Developed an e-commerce clothing website using WordPress & WooCommerce with a responsive design, implemented user authentication, product catalog, shopping cart, and QR code payment.
- Enhanced security with SSL encryption & Sucuri Security, integrated invisible watermark for content protection.

Demo Video: *https://youtu.be/zwHWVFt-Kc8*

***Link Project*:** *https://github.com/HoangCaoPhong/MAYBAO_E-commerce-clothing-website*

SKILLS

Programming Languages: Python, C/C++, LaTeX, MATLAB, C#, HTML, JavaScript, CSS.
ML/DL: PyTorch, CNNs, Transformers, Transfer Learning, Model Evaluation & Analysis, LLM/SLM Inference.
CV/NLP: OpenCV, OCR, FER, NER, ASR, Embeddings, Image Understanding, RAG-based Pipelines.
Frameworks/Tools: Git/GitHub, Docker, Apptainer, Unity, Notion, Google Workspace, Microsoft Office, WordPress.
MLOps/DevOps: CI/CD, Containerized Deployment, Model Serving, API integration, Local AI model deployment.
HPC/Parallel Computing: OpenMP, CUDA basics, GPU Computing, Parallel Programming, Performance Optimization.
Soft Skills: Leadership, Teamwork, Problem Solving, Communication, Time Management, Self-directed learning.
Languages: English (*English Intensive Program*), Vietnamese (*native speaker*), Chinese (*beginner*).